1 Solution — GIAM 1.4 Problem 5

1.1 Problem

State a theorem about the octal representation of even numbers.

1.2 The Theorem

In the style of the section's Theorems 1.4.1 (decimal) and 1.4.2 (binary):

> ***Theorem.*** *An integer is even if and only if the units digit of its octal (base-8) representation is one of $0, 2, 4, 6$ — that is, if and only if its last octal digit is itself even.*

1.3 Proof

Write an integer n in octal as

$$n \;=\; d_k\, 8^k + \cdots + d_2\, 8^2 + d_1\, 8^1 + d_0\, 8^0,$$

where d_0 is the units digit. Separate the units digit from the rest:

$$n - d_0 \;=\; d_k\, 8^k + \cdots + d_1\, 8 \;=\; 8 \cdot \big(d_k\, 8^{k-1} + \cdots + d_1\big).$$

The right-hand side is a multiple of 8, hence even. So n and d_0 differ by an even number — they have the **same parity**:

$$n \;\equiv\; d_0 \pmod{2}.$$

Therefore n is even exactly when d_0 is even. Among the octal digits $0, 1, \ldots, 7$, the even ones are $0, 2, 4, 6$. $\blacksquare$

1.4 Examples

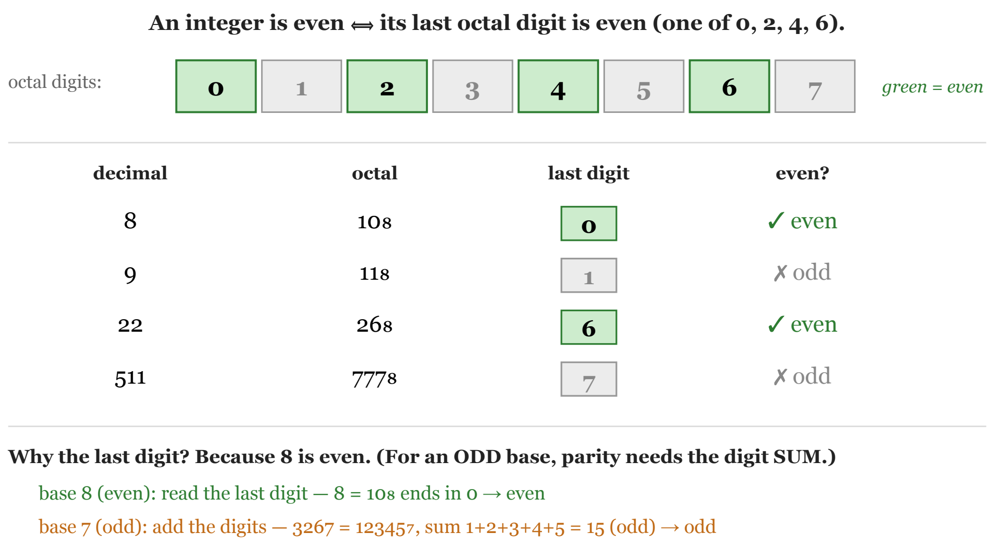

Figure 1.1: The octal digits 0–7 with the even ones (0, 2, 4, 6) marked green. Examples: $8 = 10_8$ (ends 0 → even), $9 = 11_8$ (ends 1 → odd), $22 = 26_8$ (ends 6 → even), $511 = 777_8$ (ends 7 → odd). Bottom: base 8 is even so read the last digit; an odd base like 7 needs the digit sum instead.

decimal	octal	last digit	even?
8	10_8	0	even
9	11_8	1	odd
22	26_8	6	even
511	777_8	7	odd

Table 1.1

Each verdict matches the decimal number's actual parity.

1.5 Remark — One Theorem, Three Bases

This is the octal member of a family the section has been building:

base	"even" iff the units digit is...
10 (decimal)	$0, 2, 4, 6, 8$ — Theorem 1.4.1

2 (binary)	0 — Theorem 1.4.2
8 (octal)	$0, 2, 4, 6$ — this theorem

Table 1.2

All three are the *same* statement, because all three bases ($10, 2, 8$) are **even**. Whenever the base b is even, every higher place value b^i ($i \geq 1$) is even, so only the units digit can affect parity. The even digits available shrink with the base: $\{0, 2, 4, 6, 8\}$ in decimal, $\{0, 2, 4, 6\}$ in octal, $\{0\}$ in binary.

1.6 Remark — Odd Bases Behave Completely Differently

The last-digit test relies on the base being even. In an **odd** base the test *fails*, and parity is governed by the **digit sum** instead. The reason: if b is odd then $b \equiv 1 \pmod 2$, so $b^i \equiv 1 \pmod 2$ for every i, and

$$n = \sum_i d_i b^i \equiv \sum_i d_i \pmod 2.$$

So in an odd base, n is even iff the *sum of its digits* is even. Base 7 (from Problem 4) is the perfect test case: $3267 = 12345_7$, whose digit sum is $1 + 2 + 3 + 4 + 5 = 15$ — *odd* — correctly predicting that 3267 is odd (it ends in 7 in decimal). The *last* base-7 digit alone (5) would give no reliable parity information; you must add them all.

1.7 Remark — The General Divisibility Principle

The octal theorem is one instance of a single unifying fact. For a modulus m:

- **If** $m \mid b$ (the modulus divides the base): then $b^i \equiv 0 \pmod m$ for $i \geq 1$, so $n \equiv d_0 \pmod m$ — a **last-digit test**.
- **If** $b \equiv 1 \pmod m$: then $b^i \equiv 1 \pmod m$, so $n \equiv \sum_i d_i \pmod m$ — a **digit-sum test**.

This explains *every* elementary divisibility rule at once. In decimal ($b = 10$): $2, 5, 10$ divide 10, giving last-digit tests; $3, 9$ satisfy $10 \equiv 1$, giving digit-sum tests. In octal ($b =$

er is divisible by 7 iff its octal digit sum is). Our evenness theorem is just the case $m = 2$, $b = 8$.

1.8 Remark — Doubly-Even in Octal (Tying Back to Problems 1–3)

The same idea extends the parity test to higher powers of 2, connecting to the "doubly-even" thread. Since $8 \equiv 0 \pmod 4$, we have $8^i \equiv 0 \pmod 4$ for $i \geq 1$, so $n \equiv d_0 \pmod 4$. Hence:

> *An integer is **doubly-even** (divisible by 4) iff its last octal digit is 0 or 4.*

For example $12 = 14_8$ ends in 4, and indeed $4 \mid 12$; $8 = 10_8$ ends in 0, and $4 \mid 8$; but $22 = 26_8$ ends in 6 (not 0 or 4), so 22 is *not* doubly-even — consistent with Problem 1. And since $8 \equiv 0 \pmod 8$ too, divisibility by 8 (triply-even) is read off the last octal digit being exactly 0. So octal makes the whole "k-times even" hierarchy visible in the units digit — the base-8 counterpart of the trailing-zeros test that binary gave in Problem 2.